1D CNN — Two-Hand Dynamic Gesture Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic · 31 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic · 24 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic · 5 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic · 30 trials
Total training paths	7 · 180 trials total
Batch-test data root	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Saved model path	saved_models/cnn_model_2026-05-07_22-53-04_cf3b42f32067.keras
Experiment log	saved_models/experiment_results.csv

Sensor selection

Hands	left, right
Segments	wrist, palm, thumb, index, middle, ring, pinky
Features	YPR, Accel, Flex
Channels	176 (sequence length 90)

Preprocessing & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Test size / seed	0.2 · random_state=42 · stratify=True
Sample counts	total=180 · train pool=144 · train+aug=432 · hold-out=36
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Augmentation (training only)

Enabled	yes
Copies per sample	2
Random seed	42
Active augmentations	amplitude_scale, baseline_offset
• amplitude_scale	apply_prob=0.5, scale_range=(0.95, 1.05)
• baseline_offset	apply_prob=0.5, offset_std_ratio=0.02

Model architecture & training

Architecture	Conv1D(32, k=4, ReLU) → MaxPool(2) → Conv1D(64, k=4, ReLU) → MaxPool(2) → Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(softmax)
Optimizer / loss	adam · sparse_categorical_crossentropy
Input shape	(90, 176)
Trainable params	113,190
Epochs · batch size · val split	15 · 16 · 0.2

Cross-validation

Folds · shuffle	5 · True
Mean ± std val acc	0.9653 ± 0.0378
Best fold	2

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	345	29	0.8966	1.0000	0.8966
2	345	29	1.0000	1.0000	1.0000
3	345	29	0.9655	1.0000	0.9655
4	345	29	1.0000	1.0000	1.0000
5	348	28	0.9643	1.0000	0.9643

Final training & hold-out test

Final train acc / loss	0.9971 · 0.0224
Final val acc / loss	1.0000 · 0.0209
Best val acc / lowest val loss	1.0000 · 0.0192
Hold-out test acc / loss	1.0000 · 0.0090

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	6

Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	6
Double_cmere	1.000	1.000	1.000	7
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Batch test (NewTestData)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	**20**	**30**	**0.667**